Carl von Ossietzky Universität Oldenburg

Introduction to psy126

Creating your own Jupiter Notebook

Leonardo Zaggia
10.04.26

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Oldenburg

An introduction to psy126 seminar

- Module structure
- Course info
- Semester schedule
- The Portfolio – exam dates and protocol
- Setting up
- Markdown syntax

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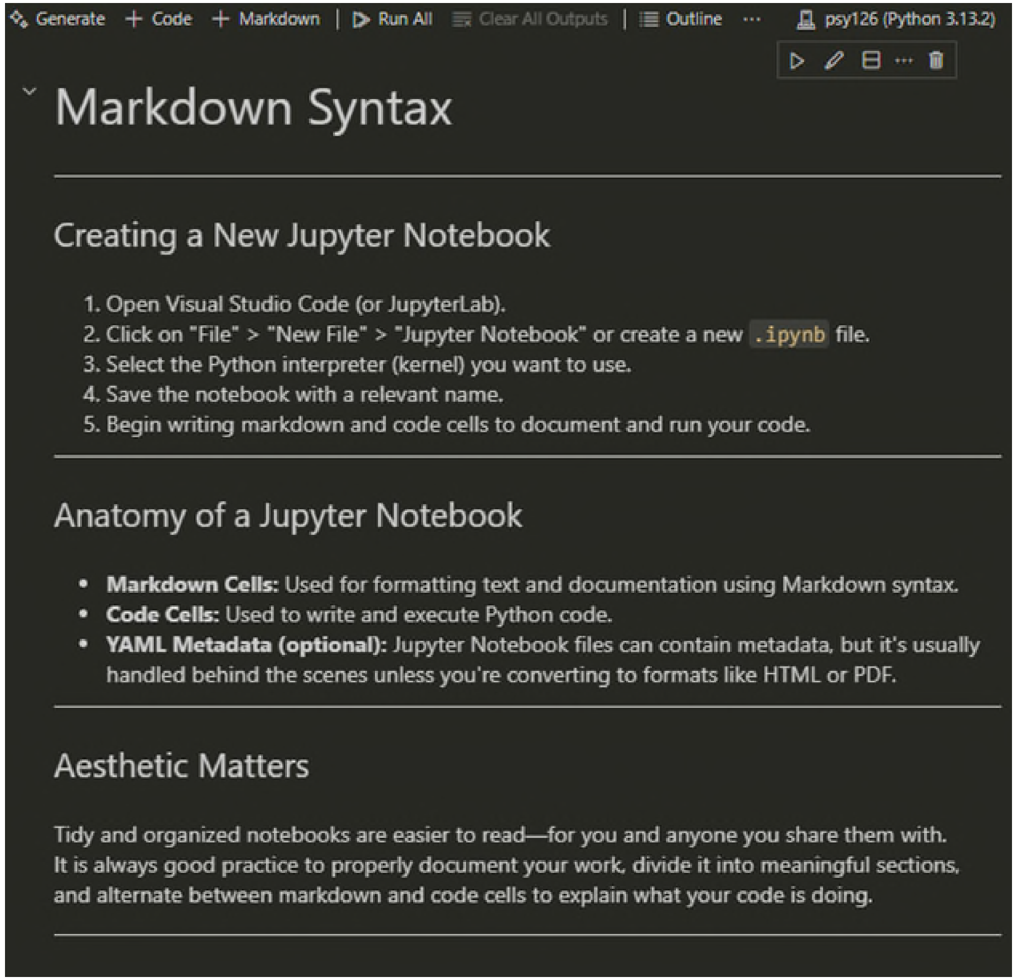
Overview – Create a .IPYNB file and leveraging its markdown features



Generate + Code + Markdown | Run All Clear All Outputs | Outline ··· psy126 (Python 3.13.2)

Markdown Syntax

Creating a New Jupyter Notebook

1. Open Visual Studio Code (or JupyterLab).
2. Click on "File" > "New File" > "Jupyter Notebook" or create a new `.ipynb` file.
3. Select the Python interpreter (kernel) you want to use.
4. Save the notebook with a relevant name.
5. Begin writing markdown and code cells to document and run your code.

Anatomy of a Jupyter Notebook

- **Markdown Cells:** Used for formatting text and documentation using Markdown syntax.
- **Code Cells:** Used to write and execute Python code.
- **YAML Metadata (optional):** Jupyter Notebook files can contain metadata, but it's usually handled behind the scenes unless you're converting to formats like HTML or PDF.

Aesthetic Matters

Tidy and organized notebooks are easier to read—for you and anyone you share them with. It is always good practice to properly document your work, divide it into meaningful sections, and alternate between markdown and code cells to explain what your code is doing.

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psy126 module structure

Lecture

Test Theory and Test Construction
with Andrea Hildebrandt, Tuesdays 2-4 pm

Seminar

Test Analysis Applied
with Leonardo Zaggia, Fridays 8-10 am

Introduction – Test analysis applied

- Work opportunities

- Foundational to the most remarkable achievement in psychology

- Acquire critical thinking on controversial and crucial topics
  - Validity and fairness
  - Standardized testing bias
  - Impact on educational policy

Introduction – Test analysis applied

- Work opportunities

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(You want to get a good grade on the Report)

Course info – portfolio evaluation criteria

- Attend at least 70% of the Seminar – but do not forget the Lectures!
- Follow the structure of the template
- Code along the classes
- Ask many questions – use the forum section of the course!
- Use the material that will be provided in the Lecture

Course info – Administrative

- We will check the attendance at the beginning of the lecture
- Strict 70% attendance -> do not strategically skip classes
- Registration for exam: 18th May – 30th June
- Deadline for submission is the 30th of September
- You will not be able to withdraw or register after 30th of June 2026
- Submit only one .ipynb project -> more on submission in later sections

Content of the lecture

- 07.04. - Introduction to Psychometrics
- 14.04. - Recap: Classical test theory
- 21.04. - Generalizability theory
- 28.04. - Latent state and trait theory
- 05.05. - Measurement models for dichotomous scores
- 12.05. - Measurement models for polytomous scores
- 19.05. - Measurement models for quantitative scores
- 26.05. - Measurement invariance / differential item functioning
- 02.06. - Measurement invariance across time
- 09.06. - Multidimensional models
- 16.06. - Preference modeling (forced choice items against faking)
- 23.06. - Network models for clinical symptom measurements
- 30.06. - Machine learning in Psychometrics
- 07.07. - Summary

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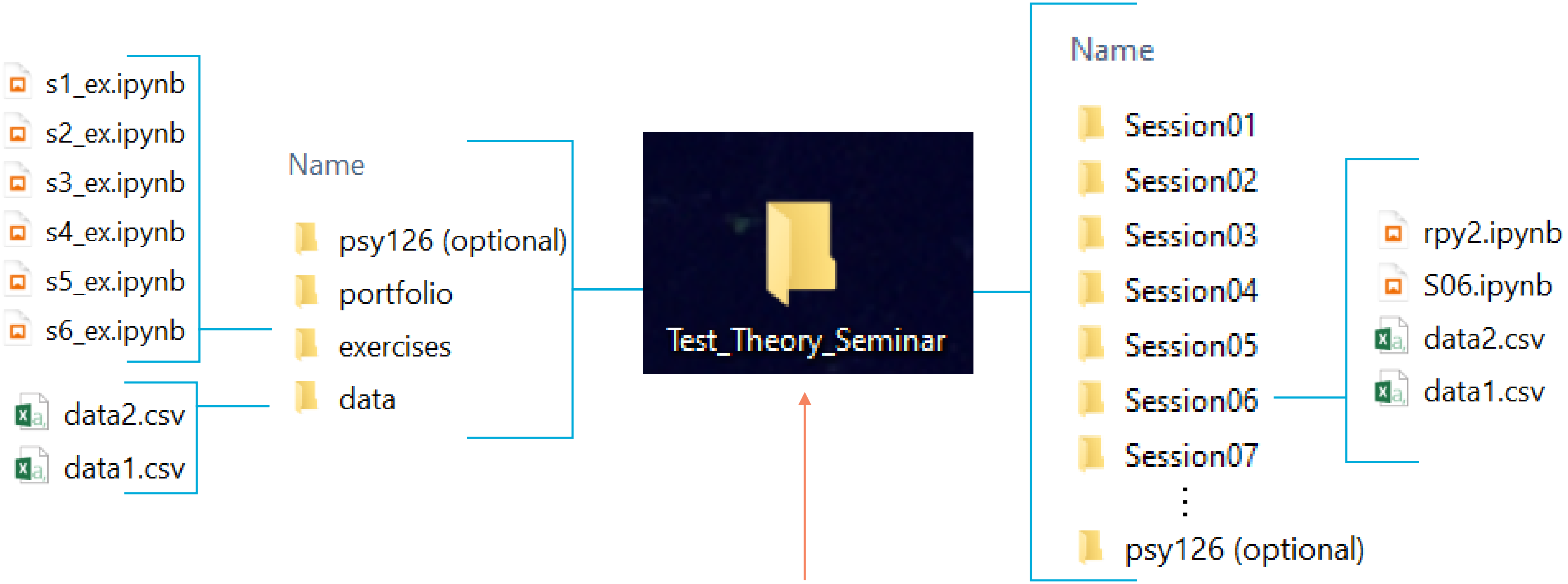
Content of the seminar

- 10.04. - Introduction to creating your own Jupiter Notebook
- 17.04. - Recap psy111 and import / export data in Python
- 24.04. - Distribution of the tasks for portfolio
- 01.05. - Public holiday
- 08.05. - Import simulated data for your portfolio and explore the data
- 15.05. - Import simulated data for your portfolio and explore the data
- 22.05. - Apply measurement models for dichotomous scores
- 29.05. - Apply measurement models for polytomous scores
- 05.06. - Apply measurement models for quantitative scores
- 12.06. - Apply measurement invariance analysis (groups, time)
- 19.06. - Apply your own analysis
- 26.06. - Apply multidimensional models for quantitative data
- 03.07. - Apply your own analysis
- 10.07. - Questions Answers for completing your portfolio

First steps – Folder structure



Be sure this folder is stored locally and not on cloud based partitionings such as OneDrive or iCloud

(R) you ready for..